

SAATHI — Product Requirements Document

Implementation-ready PRD for OpenCode • source-derived from the supplied SAATHI PRD, TRD, UI reference and verified ML artifacts

1. Product Definition

SAATHI is a private victim-support web application combining a case workspace, one-time sensitive profile setup, periodic open-ended well-being check-ins, a local machine-learning signal, optional Gemini conversational support, and an administration queue for authorised follow-up.

SAATHI is *not* an emergency-response platform, hospital system, legal decision engine, or clinical diagnostic system. Emergency support is a bypass route to human help. AI provides support signals and routing assistance; humans make consequential decisions.

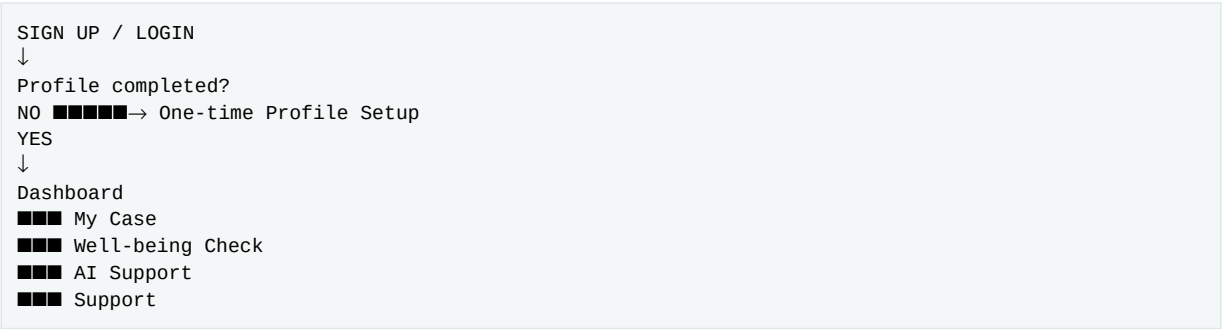
2. Target Users

Role	Primary capabilities	Boundary
Victim / User	Own profile, case, check-ins, support and AI support	Own records only
Psychologist / Counsellor	Assigned well-being cases and authorised support information	Assigned scope only
District Officer	Authorised case records, service requests and administrative actions	Authorised district scope
State / National Administrator	Aggregated programme monitoring and analytics	No individual sensitive details by default

3. Product Goals

- Give a vulnerable user one safe place for their case, support and check-ins.
- Collect the sensitive profile exactly once after first successful login/signup.
- After successful case ingestion, replace the ingestion UI with the case-details workspace.
- Run the supplied local TF-IDF + Logistic Regression model on open-ended check-in text.
- Use Gemini for conversational support and optional document understanding, never as the primary well-being classifier.
- Create a simple, explainable administrative priority queue.
- Keep the MVP as one FastAPI backend, one frontend and one database; do not introduce microservices, Kafka or Redis for the MVP.

4. User Journey



5. One-time Sensitive Profile

- If `profile_completed` is false, redirect to `/profile/setup`.
- Required: full name, preferred display name, phone number, preferred language, user role in case, consent acknowledgement.
- Optional only when relevant: emergency contact, safe contact method, address, case reference, relationship to victim, verified case category.
- Do not ask for unnecessary medical history, detailed trauma narrative, caste identity, sexual history or other sensitive information without a documented requirement.
- Sensitive profile values must not appear in URLs, browser `localStorage`, analytics events or frontend source code.

6. Authentication and Signup

- 1 User lands on SAATHI and chooses Log in or Sign up.
- 2 For signup, choose role first.
- 3 Enter full name, account mobile, optional email, password and confirmation.
- 4 Password is validated before account creation continues.
- 5 Create the account in an unverified state and initiate the MSG91 OTP flow.
- 6 The OTP verifies the `<b>account mobile number</b>` only.
- 7 Backend verifies the MSG91 access token before activating the account.
- 8 After verification, complete the one-time profile setup.
- 9 Emergency contact is a separate optional field and is never substituted for the account mobile.

7. Case Requirements

Before case ingestion, show four choices: Document / Photo Scan, e-Courts CNR Lookup, Voice Statement and Guided Questions. MVP upload accepts PDF, JPG and PNG, with a 10 MB maximum. A sample case is allowed only in development/demo mode.

After successful ingestion, the normal case page must show case details only: case ID/verified status, current stage, next hearing when known, authorised relief/protection status, timeline, summary, documents and authorised updates. The upload box and ingestion cards must disappear unless the user explicitly enters the update flow.

8. Well-being Check

- 1 Authenticate the user.
- 2 Validate text length.
- 3 Store the raw check-in securely.
- 4 Transform text with the supplied TF-IDF vectorizer.
- 5 Run the supplied Logistic Regression classifier.
- 6 Store class ID and confidence; expose a label only if verified training metadata exists.
- 7 Run deterministic support-routing rules separately.
- 8 Return a supportive, non-diagnostic response.

The supplied artifacts establish six numeric classes [0,1,2,3,4,5], 5,000 input features and a 5,000-entry TF-IDF vocabulary. Their semantic class names must come from the original training metadata; OpenCode must never invent them.

9. Gemini Features

- AI support chat through FastAPI → Gemini.
- Optional server-side document extraction.
- Optional well-being summary from already-authorised check-ins.
- Gemini must not produce the official distress score, administrative priority, legal outcome, compensation decision or protection decision.
- Gemini failures must degrade gracefully; core CRUD and local ML should remain usable.

10. Deterministic Administrative Priority

```
priority = 0
if verified_category in HIGH_PRIORITY_CATEGORIES: priority += 70
if protection_request_open: priority += 20
if explicit_human_help_request: priority += 15
if wellbeing_review_flag: priority += 10
priority = min(priority, 100)
```

High-priority categories are: rape or gang rape; murder, grievous hurt or arson; witness intimidation or threats; families affected by caste-based violence. These must be based on verified case attributes, not inferred from emotional text.

11. Functional Requirements

ID	Requirement	Acceptance condition
FR-01	Authentication	Signup, login and logout work; passwords are hashed.
FR-02	One-time profile	New user is forced to profile setup once; later logins skip it.
FR-03	Case ingestion	PDF/JPG/PNG can be uploaded and stored with ownership.
FR-04	Case details	After ingestion, My Case shows details and hides ingestion UI.
FR-05	Check-in	Authenticated user can submit open-ended text.
FR-06	ML inference	Backend loads both joblib artifacts once and predicts configured class metadata.
FR-07	Gemini chat	Frontend calls FastAPI; FastAPI calls Gemini; browser never sees the key.
FR-08	Admin priority	Verified high-priority categories appear ahead in the authorised queue.
FR-09	Audit	Sensitive access and admin changes are logged.
FR-10	Safety	Immediate safety action bypasses normal check-in flow and shows human-help instructions.

12. Non-functional Requirements

- Privacy first: no sensitive content in logs.
- Authorization on every profile, case and check-in endpoint.
- Normal CRUD must work without Gemini.
- One frontend, one FastAPI backend and one database for MVP.
- Responsive desktop and mobile UI.
- Keyboard navigation, visible focus, labels and adequate contrast.
- Target normal local API requests under 500 ms excluding model/Gemini calls.

13. MVP Out of Scope

- Kafka, Redis, microservices and Kubernetes.
- Real-time emergency dispatch or automatic police communication.
- Automatic legal advice or clinical diagnosis.
- Predicting future violence or criminal behaviour.
- Inferring sensitive identity or case category from emotional text.
- Production e-Courts integration without an approved API/source.

14. Definition of Done

- Fresh clone installs and runs.
- Signup → OTP → one-time profile → dashboard works.
- Case upload → case details works.
- Check-in → local ML works.
- Gemini works when configured and has a safe fallback when unavailable.
- Admin priority queue works from database facts.
- No tracked API secrets.
- No fake production data or invented ML labels.
- Security, authorization, responsive UI and critical-flow tests pass.